

Peddireddy Sushanth Reddy

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Professional Summary

AI/ML Engineering undergraduate (B.Tech CSE - AIML, 2026) with hands-on experience building and deploying production-grade intelligent systems. Co-inventor of a published Indian patent on AI-driven breast cancer detection. Proficient in Generative AI, LLMs, RAG pipelines, Agentic AI, and Computer Vision. Seeking to apply deep technical expertise to solve high-impact, real-world AI challenges in fast-paced innovation environments.

Education

Keshav Memorial Institute of Technology, Hyderabad
B.Tech in Computer Science & Engineering (AI & ML Specialisation)

Oct 2022 – Jun 2026
CGPA: 8.06 / 10.0

Experience

Centific Global Technologies
AI Model Engineering Intern

June 2026 – Present

- Built a Business Analyst Agent using LLMs and RAG for business analysis workflows.
- Performed LLM evaluations to assess response quality, relevance, and faithfulness.
- Developed RAG evaluation pipelines using DeepEval with curated test cases and golden answers.
- Used Python and AI/ML frameworks for evaluation, experimentation, and performance optimization.

Technical Skills

AI / ML: Machine Learning, Deep Learning, Transformers, LLMs, NLP, Generative AI, RAG, Agentic AI, MCP, Computer Vision

Frameworks & Models: EfficientNet, U-Net, U-NetR, ViT-B16, PyTorch, TensorFlow

Programming Languages: Python, Java, C++, C, JavaScript, SQL

Tech Stack: React.js, Node.js, Express, Flask, Docker, AWS, Unreal Engine, Git, GitHub

Databases: MongoDB, SQL

Soft Skills: Leadership, Teamwork, Technical Communication, Rapid Prototyping

Projects

Agentic Video RAG System with MCP [GitHub]

2026

- Built an end-to-end Video Question Answering system using Retrieval-Augmented Generation (RAG) with timestamp-based retrieval for precise video navigation.
- Integrated Ragie API for audio-video transcription, vector embedding creation, and semantic search over video content.
- Implemented MCP tools for ingestion, semantic search, and automated video clip extraction using MoviePy.
- Designed an agentic workflow enabling the LLM to autonomously select and call tools based on natural language user intent.
- Stack:** *Python, MCP, RAG, LLMs, Ragie API, MoviePy, Agentic AI*

GenAI-Powered Web App for 3D Medical Imaging [GitHub]

2025

- Developed a GenAI-based web application for automated lung and liver segmentation and nodule classification from medical scans.
- Integrated Large Language Models (LLMs) with deep learning pipelines to generate natural language radiology reports from imaging results.
- Deployed on AWS with Docker for scalable, real-time clinical decision support in production environments.
- Stack:** *MERN, Flask, Docker, AWS, GenAI, LLMs, Deep Learning, Unreal Engine*

Breast Cancer Detection Using AI & VR [GitHub]

2024

- Led a 5-member team to build an AI system detecting breast cancer in ultrasound images, integrated with immersive Virtual Reality for enhanced diagnosis.

- Achieved AI-based cancer classification and tumour segmentation using EfficientNet, U-Net, U-NetR, and ViT-B16 architectures.
- Rendered segmentation results in Unreal Engine VR for 3D tumour visualisation, improving radiologist interpretability.
- Directly resulted in a published Indian Patent (Application No. 202541059395 A).
- **Stack:** *MERN, Flask, Unreal Engine, EfficientNet, U-Net, U-NetR, ViT-B16*

Patent & Certifications

Indian Patent: 202541059395 A

Published: 27 Jun 2025

- Co-inventor of “*Classification & Segmentation of Breast Cancer Mass in Ultrasound Imagery/Video with 3D Visualisation*”, filed in association with KMIT.
- Filed 20 June 2025 under Application No. 202541059395 A, India. Patent work is based on the Breast Cancer Detection Using AI & VR project.

Achievements & Awards

- Winner at Centific Premier Hackathon 2.0 in AI COE Stream
- Top Performer – KMIT Internal Hackathon for Smart India Hackathon (SIH)
- National Level Badminton Player
- Olympiad Winner: SIMO, SIPHO, SICHQ, Ramanujan Mathematics Olympiad

Leadership & Engagement

- **Head of Photography Club, KMIT** – managed a 50+ member club and organised college-wide events
- Student Council Member, KMIT
- Member, Rotaract Club